

SQL Analysis Report – Chinook Database

1. Database Setup

Import the Chinook database into MySQL and understand its relational structure by identifying tables, primary keys, foreign keys, and the relationships between them.

The Chinook database was successfully imported into MySQL. The database consists of 11 related tables connected through primary and foreign keys. Relationships between artists, albums, tracks, customers, invoices, and invoice lines were verified using SQL JOIN operations. Understanding these relationships provides the foundation for writing analytical SQL queries in subsequent tasks.

3. Core SQL Queries

The core SQL queries demonstrated how to retrieve, filter, sort, and summarize data within the Chinook database. The use of aggregate functions and grouping enabled meaningful insights into customer distribution and sales performance, providing a foundation for more advanced SQL analysis.

4. Advanced SQL

Advanced SQL techniques were applied to combine related tables, filter data using subqueries, and perform analytical calculations with window functions. These methods are essential for analyzing relational data and generating business insights efficiently.

5. Business Insights

Which tracks generate the highest sales revenue? This query calculates the total revenue generated by each track and displays the top 10 highest-performing tracks.

- Insight: The top 10 tracks accounted for a significant share of overall sales.
- Recommendation: Feature these high-performing tracks in promotional playlists and analyze what makes them successful to inform future marketing strategies.

Which customers have spent the most money? The query aggregates each customer's total spending and ranks them from highest to lowest.

- Insight: Helena Holy was the highest-spending customer with total purchases of 49.62. The top five customers contributed a substantial portion of total revenue.
- Recommendation: Introduce a loyalty program or personalized offers to retain high-value customers.

How has revenue changed over time? This query summarizes monthly revenue, making it easier to identify sales trends over time.

- Insight: Revenue was highest in **January/2022** at **52.62** and lowest in **November/2023** at **23.76**, indicating fluctuations in sales over time.
- Recommendation: Investigate the factors behind peak and low sales periods to improve future planning.

How many purchases has each customer made? The query calculates both the number of purchases and the total amount spent by each customer.

- Insight: Helena Holy made the highest number of purchases (7 invoices), while most customers made fewer transactions.
- Recommendation: Encourage less active customers to purchase more frequently through targeted promotions.

SQL was used to answer key business questions related to product performance, customer value, revenue trends, and purchasing behavior. Additionally, indexing and query analysis techniques were introduced to improve query execution efficiency. These practices demonstrate how SQL supports both business decision-making and database performance optimization.